

7.15 block 07b split receipt — 2026-09-26

Decision and provenance

The owner authorized three stacked replacement review units after all three Copilot rounds on firmware PR #863 returned findings. PR #863 remains open as the historical complete candidate. Its final code head is 729ae7cfacf581ce534a98f6f5e0cc39efee3838, based on 8c654bcd8e1c8936a86aaa625a4ab78900a250e9. The reviews were 5323189342 (four findings), 5323944293 (two) and 5324788734 (one). All seven threads received source dispositions and are resolved. No fourth review is requested on the unsplit PR.

The #863 audit report and PDF were frozen before this authorization. Their "split pending" statement is a historical status; this receipt records the subsequent owner decision and replacement stack.

Unit	Boundary	Local code head before this receipt	Python companion
A-C formatters and interpolated intent	7a base	36dccf491b18ee50080f02e47dff49778815c32	f813519c7c1163783b0e89b5cbd70536831f5e49
D groups, one-array iteration, optional visibility	A-C	d7d12b2228231045e00bd187eb5635d5e69a393b	8fc1c2bc038db8ee2c192b162c3308234117275e
E embedded calls and assembled audit	D	f1293bc67	67e891214086f7c7c6ebc7f75f9e3e9f27ec96a0

The E firmware source matches #863's tracked tree except for the Python gitlink and this receipt. The E companion's tracked tree matches #228's df376eec6ea7d705f7a70bdd31210f8cf929f97e; its new SHA gives it a reviewable diff against the D companion. Both equivalences were checked with `git diff --exit-code` before adding this receipt. Each unit adds only the source it compiles or dispatches; D and E source is absent from the A-C unit, and E source is absent from D.

File placement

Each cell identifies the adjacent branch diff that owns that file's edits. For a shared file, edits introduced in A-C, D and E are distinct changes; later edits to an earlier line belong to the later diff. The three adjacent diffs account for the entire original #863 tracked change set. The Python gitlink identifies three companion branches with the same boundaries.

Tracked path	A-C	D	E
deps/python-keepkey	✓	✓	✓
docs/release/audit-units/715-07b-formatters-audit-20260925.md			✓
docs/release/audit-units/715-07b-formatters-audit-20260925.pdf			✓
docs/security/HANDOFF-ERC7730-715-FORMATTERS.md	✓	✓	✓
docs/security/HANDOFF-ERC7730-PHASE-E.md			✓
include/keepkey/firmware/eip712_stream.h	✓		
include/keepkey/firmware/erc7730_abi_stream.h	✓		✓
include/keepkey/firmware/erc7730_capabilities.h	✓	✓	✓
include/keepkey/firmware/erc7730_catalog.h	✓	✓	✓

include/keepkey/firmware/erc7730_field.h	✓		✓
include/keepkey/firmware/erc7730_format.h	✓		
include/keepkey/firmware/erc7730_program.h	✓		
include/keepkey/firmware/erc7730_workflow.h	✓	✓	✓
include/keepkey/firmware/ethereum.h	✓		
lib/firmware/CMakeLists.txt	✓		
lib/firmware/eip712_stream.c	✓		
lib/firmware/erc7730_abi_stream.c	✓	✓	✓
lib/firmware/erc7730_capabilities.c	✓	✓	✓
lib/firmware/erc7730_catalog.c	✓	✓	✓
lib/firmware/erc7730_field.c	✓		✓
lib/firmware/erc7730_format.c	✓		
lib/firmware/erc7730_program.c	✓		
lib/firmware/erc7730_workflow.c	✓	✓	✓
lib/firmware/ethereum.c	✓		
lib/firmware/fsm.c			✓
lib/firmware/fsm_msg_ethereum.h	✓	✓	✓
scripts/emulator/test_stack07_regressions.py	✓	✓	✓
unittests/firmware/CMakeLists.txt	✓		
unittests/firmware/erc7730_abi_stream.cpp	✓		
unittests/firmware/erc7730_catalog.cpp	✓	✓	✓
unittests/firmware/erc7730_field.cpp	✓		✓
unittests/firmware/erc7730_format.cpp	✓		
unittests/firmware/erc7730_program.cpp	✓		
unittests/firmware/erc7730_workflow.cpp			✓

Finding placement and disposition

Finding group	Unit	Disposition
D1 numeric signer constants; D3 signer text and decimals; D4 native Wanchain state; D5 token table precedence; D6 intent titles; D7 line splitting; D8 control bytes and oversized raw values; D12 root signer path; D16 signer marks and raw unit	A-C	Preload refusal or exact on-screen disclosure, with native and wire controls.

D10 parallel array pairing; D18 fixed indices into dynamic arrays	D	Bounds failures remain fail closed and are documented with wire cases.
D2 embedded intent; D9 literal callee; D17 refused inner definition; inner @.value without amountPath; depth-two and call binding	E	Refuse at preload or show blind, with inner-call and binding controls.
D11 mirror limits	A-C, D, E	Each companion branch mirrors the capabilities introduced in its matching firmware unit.
D14 field title; D15 pager continuation test handling	D, E	D owns the numbered iteration title; E owns the final runtime test filtering.
Copilot round 1: scalar formatter repeated in an iteration	D	Value path must walk the active array; positive/negative firmware and mirror tests.
Copilot round 1: stale phase counts, E status and EX test count	E	Handoffs and release report corrected.
Copilot round 2: display frames overwrote path classes	D	Dedicated <code>display_frames</code> , with a class-preservation test.
Copilot round 2: replay string reader accepted NUL	A-C	Independent reader rejects NUL, with a negative control.
Copilot round 3: signer native alias hid the token address	A-C	Signer alias and address precede the amount; maximum and Wanchain boundaries tested.

The original 102-agent audit recorded 42 findings only as clusters D1–D19. The raw per-finding manifest was not retained. D13 and D19 were not named in the surviving handoff; this receipt cannot reconstruct or assign individual raw findings from those clusters. The 14 later re-audit defects are recorded in the #863 report as behaviors and controls, not as a raw manifest. These historical evidence gaps remain explicit; no finding is marked clean merely because its original row is unavailable. The seven Copilot comments are individually assigned above and resolved on #863.

Validation ledger

Unit	Local native evidence	Host/device compiler evidence	Other gates
A-C	firmware-unit 649/649 in a clean run directory; catalog, formatter, raw-overflow and replay controls	Two focused compiler/validator cases pass	ARM, full wire and hosted CI pending on split head.
D	firmware-unit 652/652; iteration and display-frame controls pass	Seven focused compiler/validator cases pass	ARM, full wire and hosted CI pending on split head.
E	firmware-unit 656/656; tree-equivalent to #863 code	Compiler/catalog 23 pass, 2 skip locally without the official registry	#863 exact-head CI 36263598278 passed 16 required jobs; split-head CI pending.

The canonical 7b pre-push script fails on inherited duplicate `nanopb` options, an absent `report-gate` test in the 7a base, and its own absent `.cppcheck-build` path. A corrected `cppcheck` invocation found no issues. The owner authorized the #863 push with this exception recorded. The split branches retain that same inherited preflight condition; each run and its result must be recorded before pushing that branch. Hosted CI and review results for replacement PRs belong to their exact heads and will be added to the PR descriptions.